

Dr Milan Župunski

Address

Djordja Servickog 1/10
21000 Novi Sad, Serbia
mzupunski@yahoo.com

Date and place of birth

December 4th, 1986, Ruma, Republic of Serbia

Previous assignments

01.07.2012 – 31.12.2013

Research trainee, Faculty of Science, University of Novi Sad, Republic of Serbia

01.01.2014 – 31.12.2019

Teaching assistant, Faculty of Science, University of Novi Sad, Republic of Serbia

01.01.2020 – 30.10.2024

Assistant Professor, Faculty of Science, University of Novi Sad, Republic of Serbia; (on leave of absence 2021–2024 for research at HHU, DE)

01.11.2021 – 30.09.2025

Researcher, Institute of Cell and Interaction Biology, HHU, Düsseldorf, Germany

01.10.2025 – 28.02.2026

Researcher, Group of Prof. Dr. Jiří Friml, ISTA, Klosterneuburg, Austria

Fellowships/Research visits

01.01.2019 – 30.04.2019

01.11.2019 – 07.03.2020

Guest Scientist in the Group of Dr. Guido Grossmann, COS, University of Heidelberg, Germany: calcium signaling in plant-environment interactions (8 months and 7 days in total).

Dissertation

2010 - 2017

Dissertation at the University of Novi Sad, under mentorship of Prof. Dr Milan Borišev
“Potential of black locust (*Robinia pseudoacacia* L. 1753) in phytoextraction of Cd, Ni and Pb”
Grade: 9.50 (out of 10 max)

Tertiary education

2005 - 2009

University of Novi Sad, Faculty of Science, Diploma in Biology

2009 - 2010

University of Novi Sad, Faculty of Science, Master in Biology

Skills

Molecular biology and biochemistry assays; Microfluidics fabrication and experimentation; Brightfield and fluorescence microscopy (epifluorescent and spinning-disk confocal); Image analysis with macro language in FIJI;

Skills (continued)

Programming language:

R

EDA tools: tidyverse

Dataviz tools: ggplot2

Programmatic reporting: RMarkdown

Exploratory and inferential statistical methods, ML (random forest), multivariate analyses, RNA-seq workflow (DESeq2, edgeR, limma).

Python

matplotlib

pandas

RNAseq analysis

Awards and Grants

An awardee of the **Travelling Fellowship** granted by **The Company of Biologists/J. Cell Science** (1901.00 EUR): research visit to Dr G. Grossmann Lab, University of Heidelberg, Germany; Oct-Dec 2019

Teaching

2012 - 2021

Teaching assistant/Dozent at University of Novi Sad, Faculty of Science, Department of biology and ecology:
Plant physiology courses for undergraduate students (lectures, course supervision, and exams)

2019

University of Heidelberg, Germany, HBIGS Core Course "Principal component analysis in Biosciences" (16 hours of teaching; 0.5 credit points)

Scientific outreach

2011-2019 Workshops for science promotion at "Night of Biology" (yearly)

Referee activity

Journals:

16 verified peer reviews; Web of Science ResearcherID: U-1696-2017.

Funding Agencies:

NCN, Poland (full length proposal)

Memberships

Serbian Society for Plant Physiology (SPPS), The Federation of European Societies of Plant Biologists (FESPB), Society for Experimental Biology (SEB).



Novi Sad, July 1 2026

(Milan Župunski)

References

Prof. Dr **Guido Grossmann** (Postdoc Advisor)
Heisenberg Professor
Heinrich-Heine-Universität Düsseldorf
Institute of Cell and Interaction Biology
Geb. 26.44 U1
Universitätsstraße 1
40225 Düsseldorf
+49 (0)211 81 14746
guido.grossmann@hhu.de

Prof. Dr. **Jiří Friml** (Postdoc Advisor)
Institute of Science and Technology (ISTA)
Am Campus 1
3400 Klosterneuburg, Austria
jiri.friml@ist.ac.at

Dr. **Tonni Grube Andersen** (Collaborator)
Max Planck Institute for Plant Breeding Research
Department of Plant Microbe Interactions
Carl-von-Linne-Weg 10
50829 Cologne, Germany
Phone: +49 221 5062 317
tandersen@mpipz.mpg.de

Prof. Dr **Ivana Maksimović** (Collaborator)
University of Novi Sad
Faculty of Agriculture
Department of Field and Vegetable Crops
Trg Dositeja Obradovića 8
21000 Novi Sad, R. of Serbia
Phone: +381214583341
ivana.m@polj.uns.ac.rs

Prof. Dr **Milan Borišev** (PhD supervisor)
University of Novi Sad
Faculty of Sciences
Department of Biology and Ecology
Trg Dositeja Obradovića 2
21000 Novi Sad, R. of Serbia
Phone: +381214852651
milan.borisev@dbe.uns.ac.rs

Selected Publications (last five years)

h-index: 18

Total citations: 1215

Metrics according to Google Scholar (July 2026).

13. Erdrich S, Keilhammer M, Sharma S, **Župunski M**, Schurr U, Grossmann G, Frunzke J, Arsova B. Phage biocontrol reduces the disease burden and modulates plant immunity through suppression of bacterial virulence. (2026). **Cell Reports**, 45(7): 117622. <https://doi.org/10.1016/j.celrep.2026.117622>

12. Babić D, **Župunski M**, Friml J. (2026). Imaging and genetic toolbox to study Arabidopsis embryogenesis. **New Phytologist**. 250: 1483-1491. [10.1111/nph.71072](https://doi.org/10.1111/nph.71072).

11. Gilliard G, Pršić J, Crowet J-M, Chemotti C, Ahmed J, Lorent J, Jolivet M-D, Egli S, Egée S, Bouyer G, Race G, van Buren L, Van Den Berghe A, Argüelles-Arias A, Mathelié-Guinlet M, Ibrahim H, Genva M, Fouillen L, Mirande-Bret C, Razin O, Vue B, **Župunski M**, Luzuriaga-Loaiza WP, Deboever E, Nasir MN, Lins L, Van Der Smissen P, Boubsi F, Eschrig S, Germain V, Höfte M, Zipfel C, Dufrêne YF, Tyteca D, Koutsoubas A, Brocca P, Grossmann G, Ranf S, Dorey S, De Coninck B, Nürnberger T, Mongrand S, Gronnier J, Rondelli V, Deleu M, Ongena M. (2026). Membrane remodelling mediates lipopeptide-induced immunity in Arabidopsis. **Nature Plants**, 12: 1034–1050. <https://doi.org/10.1038/s41477-026-02270-3>

10. Chaudhary A, Hsiao Y-C, Yeh F-L J, **Župunski M**, Zhang H, Aizezi Y, Malkovskiy A, Grossmann G, Wu H-M, Cheung A Y, Xu S-L, Wang Z-Y. (2025). FERONIA signaling maintains cell wall integrity during brassinosteroid-induced cell expansion in Arabidopsis. **Molecular Plant**, 18(4): 603-618. <https://doi.org/10.1016/j.molp.2025.02.001>

9. Santos MP, **Župunski M**, Monda H.*et al.* (2025). Humic acids modify root architecture in *Arabidopsis* through H⁺-ATPase-dependent target of rapamycin activation in concert with Ca²⁺ and ROS signaling. **Chemical and Biological Technologies in Agriculture**, 12:47. <https://doi.org/10.1186/s40538-025-00764-4>

8. Matijević I, Arsenov D, Popov N, Živkov-Baloš M, Nikolić N, Pajević S, Borišev M, **Župunski M***. (2025): Acclimatization mechanisms in *Populus deltoides* roots to zinc excess. **Acta Physiologiae Plantarum** 47: 103. <https://doi.org/10.1007/s11738-025-03850-0>

7. Joksimović A, Arsenov D, Borišev M, Djordjević A, **Župunski M***, Borišev I. (2025) Foliar application of fullerene and zinc oxide nanoparticles improves stress resilience in drought-sensitive *Arabidopsis thaliana*. **PLoS ONE**, 20, e0330022. <https://doi.org/10.1371/journal.pone.0330022>

6. Borišev M, **Župunski M***, Arsenov D, Nikolić N, Tarčak S, Pajević S. (2024). Understanding beech (*Fagus sylvatica* L.) photosynthetic responses to microhabitat water deficit: a site-specific investigation. **Eur J Forest Res**. <https://doi.org/10.1007/s10342-024-01702-z>

5. Beljin J, Arsenov D, Slijepčević N, Maletić S, Đukanović N, Chalot M, **Župunski M**, Tomašević Pilipović D (2023). Recycling of polluted dredged sediment – Building new materials for plant growing. **Waste Management**, 166: 13-24.

4. VanBuren R, Wai CM, Giarola V, **Župunski M**, Pardo J, Kalinowski M, Grossmann G, Bartels D. (2023). Core cellular and tissue-specific mechanisms enable desiccation tolerance in *Craterostigma*. **Plant Journal**, 114: 231-245. <https://doi.org/10.1111/tpj.16165>
3. Sadoine M, De Michele R, **Župunski M**, Grossmann G, Castro-Rodriguez V. (2023). Monitoring nutrients in plants with genetically encoded sensors: Achievements and perspectives, **Plant Physiology**, kiad337. <https://doi.org/10.1093/plphys/kiad337> (review article)
2. Fuchs VAF, Denninger P, **Župunski M**, Jaillais Y, Engel U, Grossmann G. (2021). Nanodomain-mediated lateral sorting drives polarization of the small GTPase ROP2 in the plasma membrane of root hair cells. **bioRxiv**, 459822. <https://doi.org/10.1101/2021.09.10.459822>
1. **Župunski M***, Arsenov D, Borišev M, Nikolić N, Pajević S. (2022). Should I GROW or should I SLOW: a meta-analysis of fast-growing tree-species grown in cadmium perturbed environment. **Physiologia Plantarum**, 174: e13594. 10.1111/ppl.13594

Conference Contributions

Župunski M. Decoding Environmental Signals: Calcium Responses to Abiotic and Biotic Stimuli in Arabidopsis Roots. The 5th International Conference on Plant Biology (24th SPPS Meeting), 2024, Srebrno jezero, Serbia. (**Talk, selected**).

* Corresponding author.